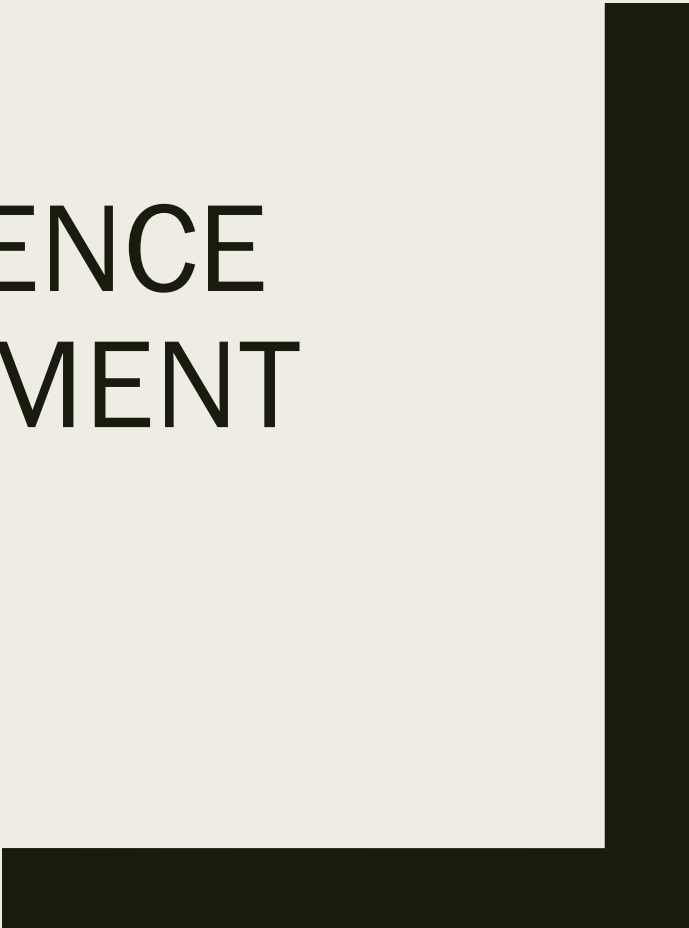




ARTIFICIAL INTELLIGENCE SOFTWARE DEVELOPMENT

Week 3 Lecture 1
Dr. Hari M Koduvely



Agenda for Today

- ❑ Theory: 5:30PM – 7:30PM
 - Recap of MLOPS from Term 1
 - Feature Engineering
- ❑ Lab: 7:30PM – 9:30PM
 - Standup Meetings

MLOps Recap

Summary of topics covered in course 1

- ☐ Data Engineering
- ☐ Training Data Generation

MLOps Recap

Summary of topics covered in course 1

❑ Data Engineering

- Data Sources
- Data Formats
- Data Models
- Modes of Dataflow

❑ Training Data Generation

MLOps Recap

Summary of topics covered in course 1

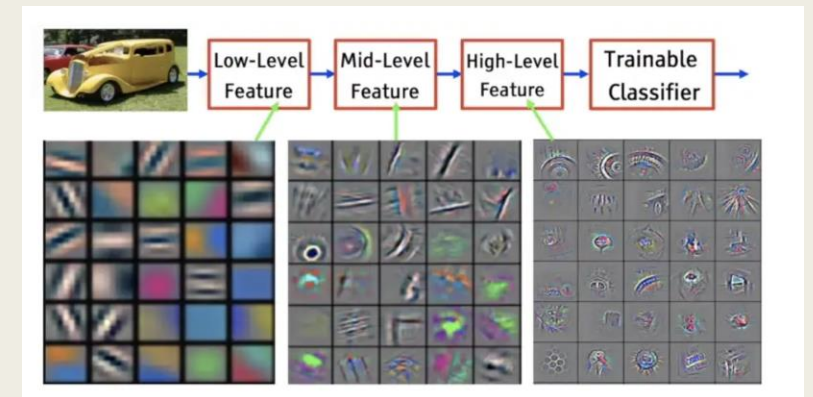
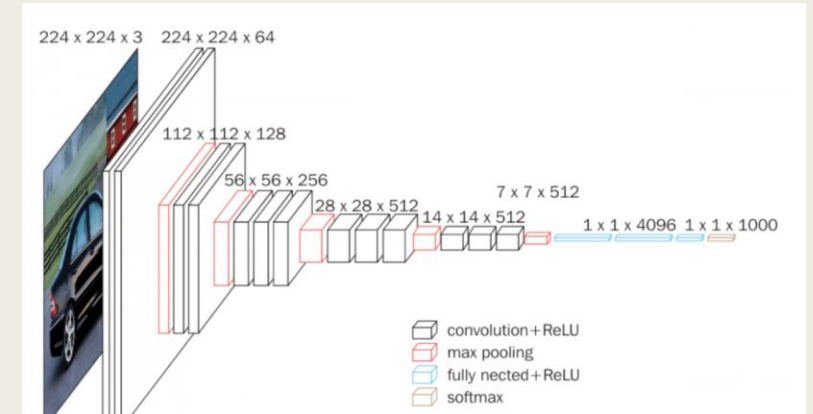
- ❑ Data Engineering
- ❑ Training Data Generation
 - Data Labelling
 - Sampling Techniques

Feature Engineering

- ❑ Importance of Feature Engineering
- ❑ Handling Missing Values
- ❑ Feature Scaling
- ❑ Encoding Categorical Features
- ❑ Positional Embeddings
- ❑ Data Leakage
- ❑ Feature Selection

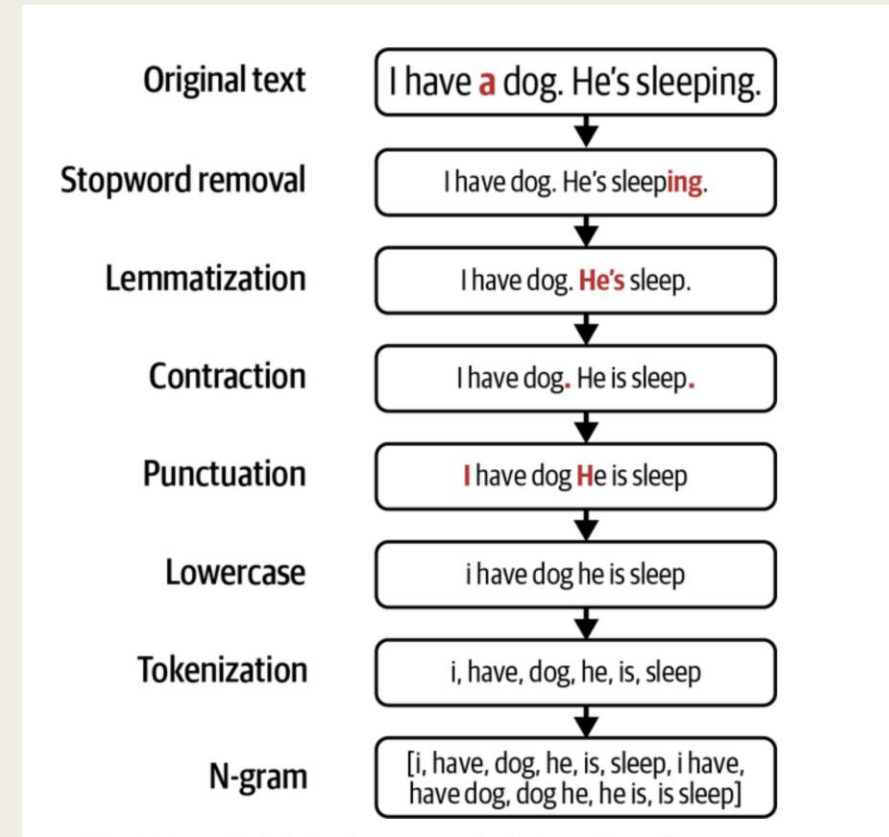
Importance of Feature Engineering

- ❑ Features are like Signals.
- ❑ Feature Engineering is like separating Signals from Noise.
- ❑ Deep Learning algorithms are capable of Learning Features themselves.



Importance of Feature Engineering

- ❑ Features are like Signals.
- ❑ Feature Engineering is like separating Signals from Noise.
- ❑ Deep Learning algorithms are capable of Learning Features themselves.
- ❑ Classical ML algorithms requires manual creation of useful features



Handling Missing Values

❏ Not all types of missing values are the same

-  **Missing Completely at Random (MCAR)**
 - The probability of a value being missing is **unrelated** to both the **observed** and **unobserved** data.
 - This means the missingness is purely random.
 - **Generally the least problematic type**, as the missing data does not introduce bias and can be safely ignored under many standard analyses (though it reduces power).
-  **Missing at Random (MAR)**
 - The probability of a value being missing is **related to other observed variables** in the dataset, but **not** to the value of the missing data itself.
 - Can be handled using statistical methods like **multiple imputation** or **maximum likelihood**, which model the observed relationships.
-  **Missing Not at Random (MNAR)**
 - The probability of a value being missing is **related to the unobserved (missing) data itself**.
 - This is the **most challenging type**, as the missingness introduces **systematic bias** and can't be addressed using only the observed data.
 - **Requires more complex statistical methods**, such as **Bayesian models**, **pattern-mixture models**, or sensitivity analyses.

ID	Age	Gender	Annual income	Marital status	Number of children	Job
1		A	150,000		1	Engi
2	27	B	50,000			Teac
3		A	100,000	Married	2	
4	40	B			2	Engi
5	35	B		Single	0	Doct
6		A	50,000		0	Teac
7	33	B	60,000	Single		Teac
8	20	B	10,000			Stud

Handling Missing Values

❑ Not all types of missing values are the same

■ ☒ 1. Missing Completely at Random (MCAR)

- **Example:**
In a survey, some respondents accidentally skip a question about their favorite color due to a printing error on some forms.
- **Key point:** The missingness is completely unrelated to any variable — observed or unobserved.
- **Effect:** No bias is introduced; the missingness is pure randomness.

■

■ ☒ 2. Missing at Random (MAR)

- **Example:**
In a medical study, younger participants are more likely to skip reporting their income. However, the participants' ages are recorded.
- **Key point:** The missingness (of income) depends on an **observed variable** (age), but not on the actual income value.
- **Effect:** Bias can be corrected using statistical methods that incorporate the observed variable (age).

■

■ ☒ 3. Missing Not at Random (MNAR)

- **Example:**
In a mental health survey, people with severe depression are less likely to answer questions about their mental state.
- **Key point:** The missingness depends on the **missing value itself** (level of depression).
- **Effect:** This introduces bias that **cannot be corrected** using just the available data. Advanced modeling or strong assumptions are needed.

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Handling Missing Values

❑ How to treat missing values in the data?

❑ Deletion:

- Column Deletion
- Row Deletion
- Easy to implement
- Can leads to accuracy loss

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Handling Missing Values

❑ How to treat missing values in the data?

❑ Imputation:

- Mean
- Median
- Mode
- Interpolation (e.g. KNN)
- Can create Bias in the data
- Can cause Data Leakage

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Feature Scaling

- ❑ Natural scale of different features are not same
- ❑ ML algorithm does not know this
- ❑ Make all features in the same numerical range
- ❑ For the range [0,1] scale factor:

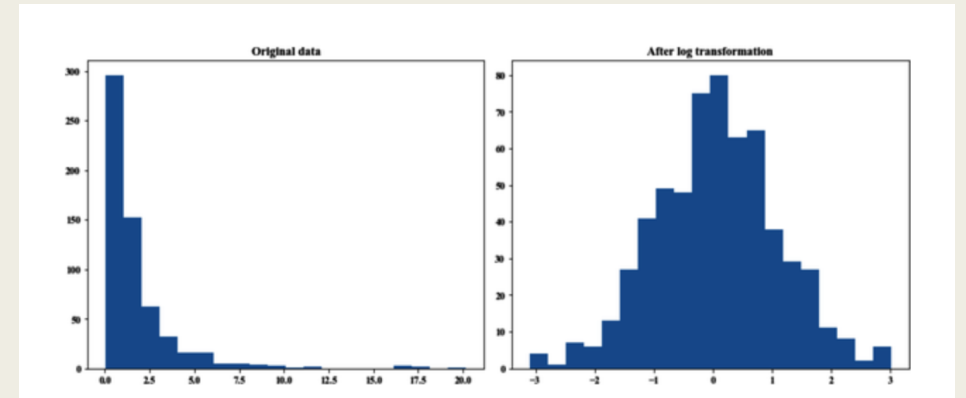
$$[x - \min(x)] / [\max(x) - \min(x)]$$

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Feature Scaling

- ❑ Not all features would be having Normal Distribution
- ❑ Transform the features to make them Normal
 - Box-Cox transformation

$$x' = (x^a - 1) / a \quad \text{for } a \neq 0$$
$$= \log(x) \quad \text{for } a = 0$$



Discretization

- ❑ Process of converting continuous features to discrete features
- ❑ Aka Quantization or Binning
- ❑ Example:
 - Create bucket for ages: 0-10, 10-18, 18-30, 30-50, 50-65, 65-80, 80+
 - Need to be careful when choosing value of the boundaries
 - Use Histogram plots

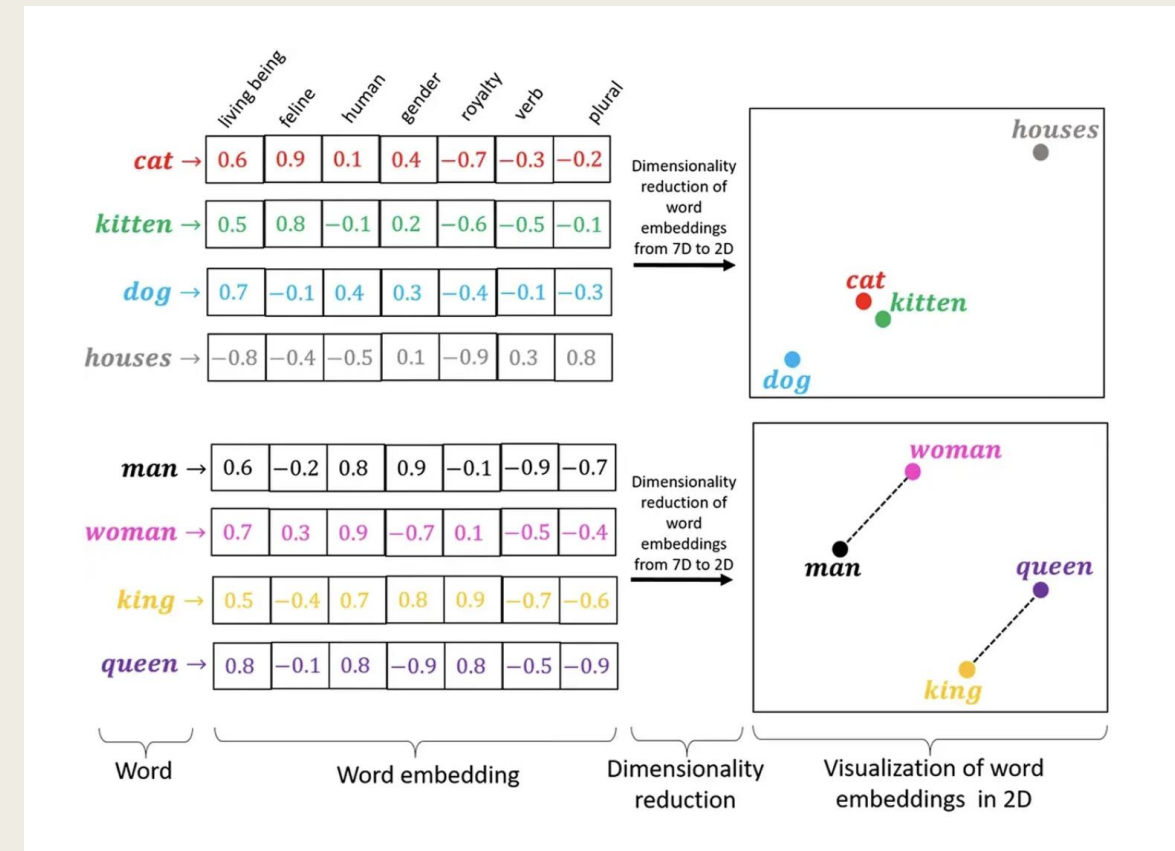
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Encoding of Categorical Features

- ❑ Some categorical features can have very large number of values
- ❑ And new values can appear in the production scenario unseen during training
- ❑ Examples:
 - IP addresses
 - Zip codes
 - Brand names

Embeddings

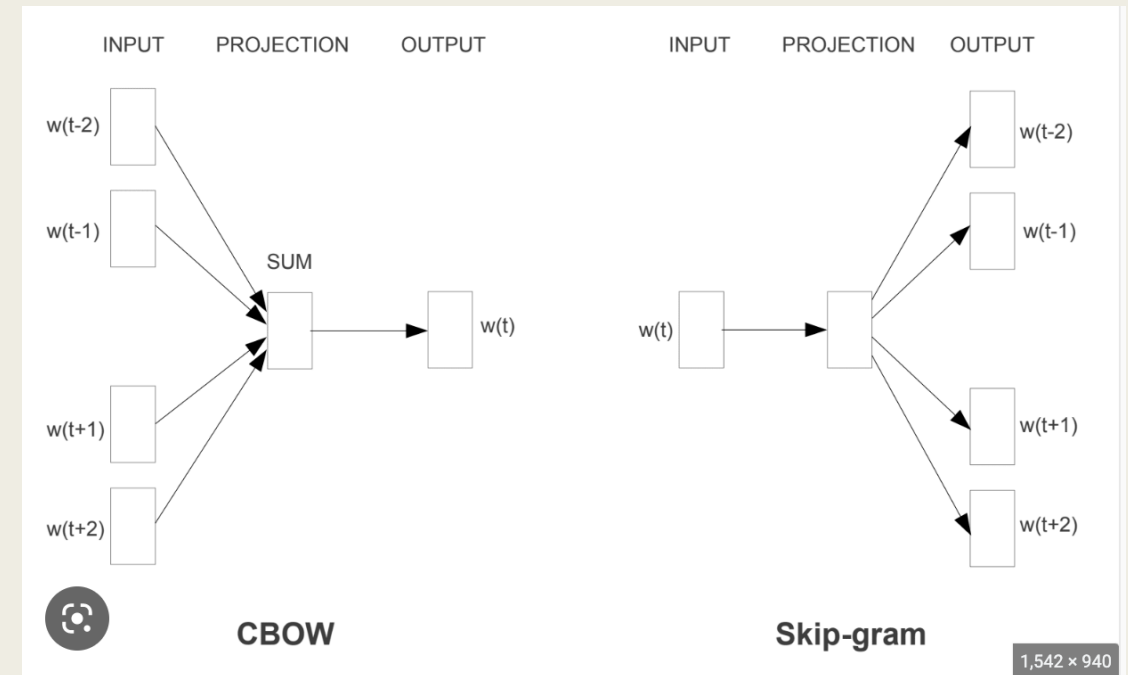
- ❑ Numerical vector representation of a categorical variable
- ❑ Words having similar semantics would be closer
- ❑ Preserves semantic relationships



Embeddings

❑ Popular word embeddings for NLP

- Word2Vec
- Glove
- Sentence Transformers



Data Leakage

- ❑ Training ML model using information not expected to be available during prediction.
- ❑ Examples:
 - **Feature Leakage** – caused by a feature which is a duplicate or proxy of the target variable.

Monthly salary as feature to predict yearly salary.
 - **Sample Leakage** – Duplicate samples between train and test datasets.
 - **Non iid Data** – Splitting a time series dataset randomly

Common Causes of Data Leakage

- ☐ Filling in missing data before splitting.
- ☐ Not removing duplicates before splitting.
- ☐ Scaling before splitting.
- ☐ Splitting time-correlated data randomly instead of by time.
- ☐ Group leakage.

How to Detect Data Leakage

- ☐ Measure the correlation between each feature and target variable
- ☐ Investigate cases of very high correlation
- ☐ Measure the temporal correlation between train and test split

Feature Selection

- ❑ Adding more features leads to better model performance.
- ❑ Having too many features can have negative impacts also:
 - More chances of data leakage
 - Could cause overfitting
 - May require more memory to serve the model
 - Could increase latency at inference

Feature Selection

- ❑ Two factors to consider while selecting a feature:
 - Importance to the model
 - Generalization to unseen data

Feature Importance

Shapley Values

- ❑ Concept borrowed from Co-operative Game Theory (1950s).
- ❑ Invented by Lloyd Shapley.
- ❑ In ML also known as SHAP (SHapley Additive exPlanations)
- ❑ Used for fairly attributing a player's contribution to the end result of a game.
- ❑ Think of ML as a co-operative game by all the features to make a prediction.

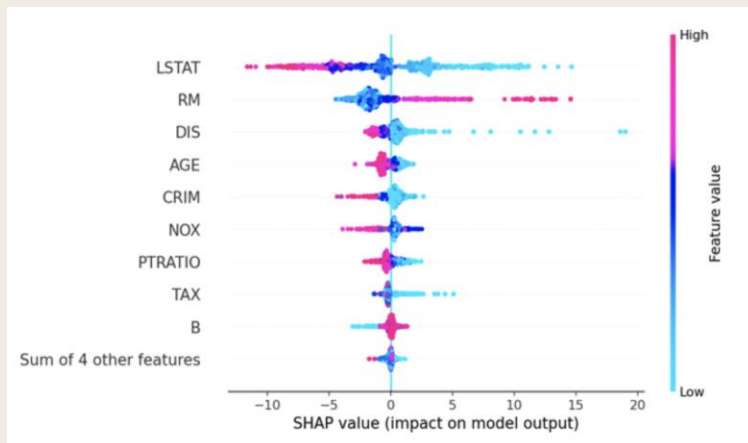
Feature Importance

Shapley Values

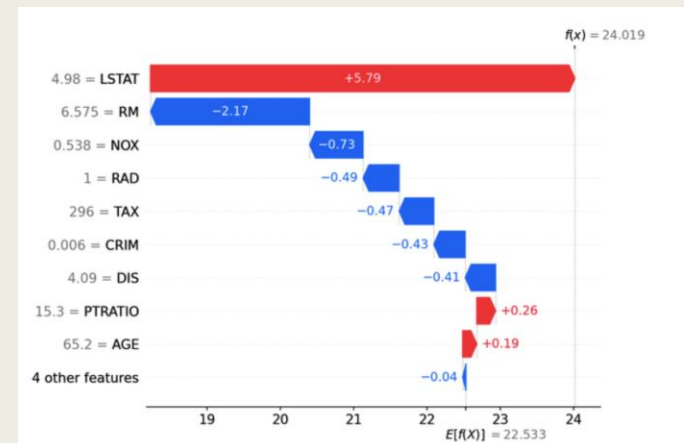
- ❑ Computed by perturbing values of input features and measuring how it is changing the model prediction.
- ❑ The Shapley value of a given feature is the average marginal contribution to the overall model score.
- ❑ Can be used for both global importance and single prediction

Feature Importance

Shapley Values



Global Feature Importance



Single Prediction Feature Importance

Feature Generalization

- ❑ ML model should make accurate predictions on unseen data
- ❑ Measuring generalization capability of features is more difficult
- ❑ Two factors to consider for feature generalization:
 - Feature coverage
 - Distribution of feature values

Summary of Best Practices

- Split data by time into train/valid/test splits instead of doing it randomly.
- If you oversample your data, do it after splitting.
- Scale and normalize your data after splitting to avoid data leakage.
- Use statistics from only the train split, instead of the entire data, to scale your features and handle missing values.
- Understand how your data is generated, collected, and processed. Involve domain experts if possible.
- Keep track of your data's lineage.
- Understand feature importance to your model.
- Use features that generalize well.
- Remove no longer useful features from your models.

Feature Importance

Shapley Values

- ❑ [Google Colab Notebook - Credit Risk Score Prediction](#)